CLINICAL · MOUTH & THROAT

Swallowed foreign bodies

A swallowed coin in a toddler, a fish bone in an adult's throat, a meat bolus in an elderly man, a denture after a seizure. Most swallowed objects pass harmlessly. The ENT surgeon's job is to find the ones that will not: those stuck in the pharynx or oesophagus, and those that are sharp, large, multiple magnets or button batteries. Remove them under direct vision before they ulcerate, perforate or block the airway. A button battery in the oesophagus is an emergency measured in hours.

AT A GLANCE

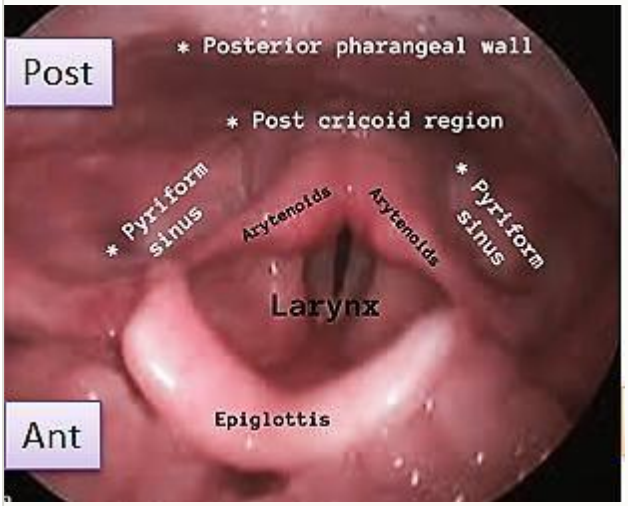
- Commonest site: just below the cricopharyngeus.
- Oesophageal coin is flat on the PA film; tracheal coin is edge-on.
- Button battery: halo sign, remove immediately, honey or sucralfate en route.
- Never use a Foley catheter or push it down blind.

1 CHAPTER ONE Where they lodge

Pharynx and oesophagus.

1.1 The pharynx

In the pharynx, fish bones and needles lodge in the **tonsil**, often in a crypt, where they can be seen and lifted out with forceps. Fish bones also catch in the **tongue base and vallecula**, where they are found with a mirror or endoscope and removed under topical anaesthesia with curved forceps; a needle buried in the tongue base may migrate and need imaging and an open pharyngotomy. Wires, pins and staples stick to the **posterior pharyngeal wall**. In the **pyriform fossa**, fish and chicken bones, needles and dentures are common; small objects can be removed in clinic, but large or impacted ones, and all objects in children, are removed under general anaesthesia.



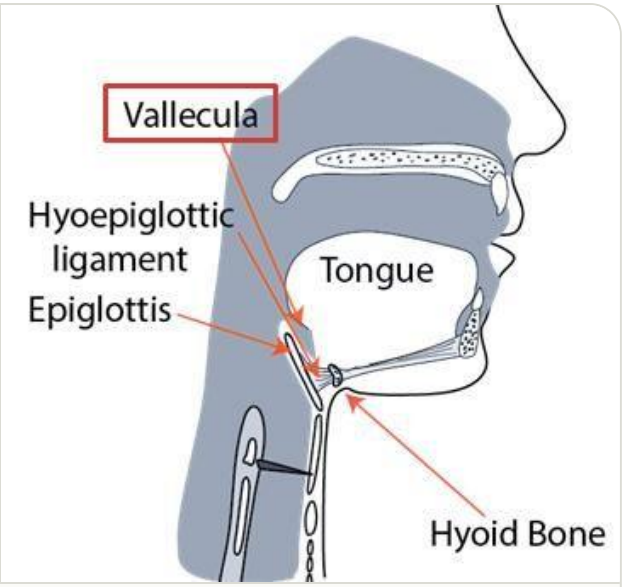


FIGURE 1 The hypopharynx and larynx as seen endoscopically, and the vallecula with the hyoepiglottic ligament.

Anatomy atlas · ENT Steps p. 221, ENT Steps p. 210

1.2 The oesophagus

By far the commonest site, about 70%, is *at or just below the cricopharyngeus*, and coins are typically held at the sphincter itself. Objects that pass it may stop at the crossing of the aorta and left bronchus (about 20%) or at the cardia (about 10%), while sharp objects catch anywhere. An object that lodges below the cricopharyngeus in an adult suggests an underlying stricture, ring, eosinophilic oesophagitis or cancer, and the oesophagus should be endoscoped or imaged once the oedema settles.

1.3 Who swallows things

About 80% of patients are children under five, who swallow coins, toys, beads, *button batteries* and magnets. Adults swallow food boluses, especially meat, and fish and chicken bones; they swallow dentures, partly because an upper denture dulls palatal sensation; and they are predisposed by hasty eating, poor chewing, alcohol, sleep, seizures and unconsciousness. Psychiatric patients and prisoners may swallow objects deliberately. In an older adult, a sudden bolus obstruction may be the first sign of a *narrowed oesophagus* from stricture or carcinoma.

2 CHAPTER TWO Assessment

History, signs, films.

2.1 Symptoms

The patient usually describes choking or gagging at the time, followed by pain or a pricking feeling localised in the neck, often beside the trachea, and worse on swallowing. Pain localises well in the pharynx and cervical oesophagus but poorly lower down. Dysphagia may begin as partial and become complete as oedema develops, and drooling with inability to swallow saliva means complete obstruction and a risk of aspiration.

In small children the respiratory tract may dominate: stridor or cough from compression of the soft posterior tracheal wall or from laryngeal oedema, or wheeze and recurrent chest infection from a long-standing oesophageal object. Chest, back or abdominal pain, fever and emphysema warn of perforation. Some patients, especially children, have few symptoms at all, so normal eating and a normal X-ray do not exclude a foreign body.

2.2 Signs

The signs are tenderness low in the neck beside the trachea (*Jackson's sign*), pain on rocking the larynx against the spine, *pooling of saliva in the pyriform fossae* that does not clear with swallowing, and sometimes the object itself in the postcricoid region. Surgical emphysema, fever and tachycardia mean perforation.

2.3 Imaging

Plain posteroanterior and lateral films are taken of the neck, chest and abdomen, and in children from mouth to anus so that every object is found. A coin in the *oesophagus* lies *flat (en face) on the PA view* and edge-on on the lateral; a coin in the *trachea* lies the other way, edge-on on the PA view, because the glottis is a sagittal slit. A button battery shows a *double ring ("halo") on the PA view* and a *step-off* on the lateral, so every apparent coin must be looked at carefully.

Many fish bones, wood, plastic, aluminium and glass are invisible on X-ray. On the lateral neck film their presence is suggested indirectly by air in the cervical oesophagus, *prevertebral soft-tissue thickening*, loss of the cervical lordosis, or air in the tissues from perforation. CT is the most sensitive test for bones, perforation and abscess, and should be used freely when symptoms persist despite a negative film. Barium is avoided because it coats the object and mucosa, can be aspirated and makes endoscopy harder; a water-soluble study is used only if perforation is suspected and CT is unavailable.

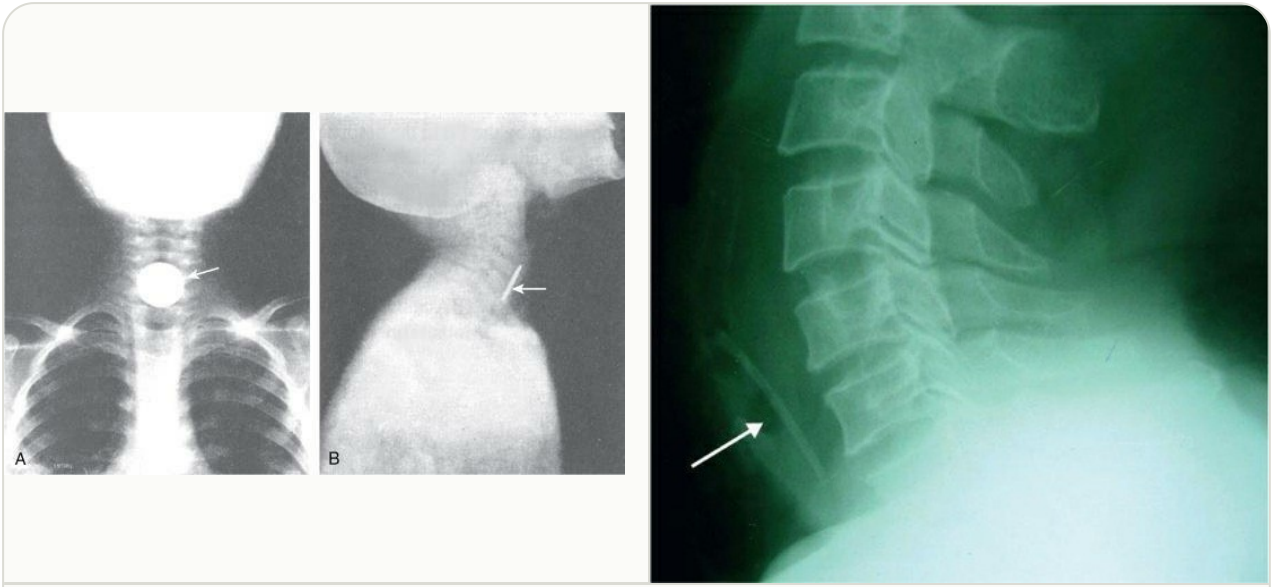


FIGURE 2 First, a coin in the upper oesophagus: (A) flat in the PA view, (B) edge-on in the lateral. Second, a fish bone in the cervical oesophagus with a bubble of air (arrow).

Dhingra, figs 70.1 and 70.2

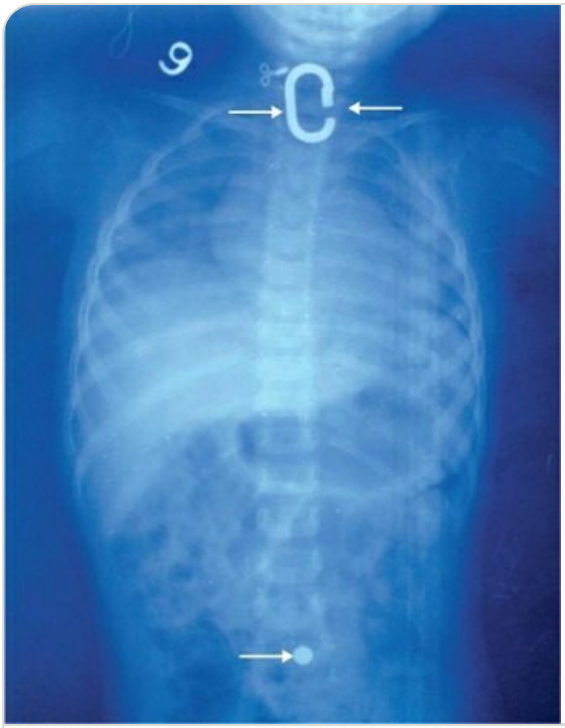


FIGURE 3 Two metallic foreign bodies: a large ring at the cricopharyngeus and a small piece in the abdomen (arrows).

Dhingra, fig. 70.3

IMAGING IN THE APP 1 case

Foreign bodies of the food and air passages on imaging.



Foreign Bodies

Trauma · 3 images

3 CHAPTER THREE Removal

Under direct vision.

3.1 Timing

Some objects must come out **within hours**: a *button battery in the oesophagus*, any object causing complete obstruction with drooling or airway compromise, sharp or pointed objects in the oesophagus, and any case with signs of perforation. Other oesophageal foreign bodies, including coins and food boluses that do not pass, should come out **within 24 hours**, because prolonged impaction increases ulceration, perforation and stricture. A smooth coin in the lower oesophagus of a well child can be **observed** for 12 to 24 hours with a repeat film, since it may pass into the stomach.

3.2 Methods

Rigid oesophagoscopy under general anaesthesia is the traditional ENT approach and the best for sharp or impacted objects and for objects at the cricopharyngeus. Sharp points are drawn into the end of the scope and withdrawn with it ("*ensheathing*"), open safety pins are closed or turned so that the point trails, and a hypopharyngoscope or long-bladed laryngoscope suits objects at the upper sphincter.

Flexible endoscopy under sedation, usually by gastroenterologists, suits food boluses, coins and blunt objects, with an overtube to protect the oesophagus from sharp ones; it also reveals any underlying eosinophilic oesophagitis, stricture or cancer, which should be biopsied. **Open surgery** is needed occasionally: a *cervical oesophagotomy* for impacted hooked objects such as partial dentures above the thoracic inlet, and *thoracotomy or thoracoscopy* for impacted lower objects and perforations.

OPERATIVE GUIDES IN THEATRE 1 operation

Oesophagoscopy and hypopharyngoscopy.

Esophagoscopy and Hypopharyngoscopy
Oesophagoscopy

METHODS TO AVOID

- Blind removal with a Foley balloon catheter: the object can be aspirated as it is pulled into the pharynx.
- Pushing an object into the stomach blind.
- Meat tenderisers (papain): they can digest the oesophageal wall.
- Emetics.
- Glucagon, which relaxes the LES but does not help when there is a stricture or ring, and has little proven benefit.

3.3 After removal

After removal the mucosa is re-examined for ulceration, perforation and an underlying cause. Oral fluids start if the procedure was uneventful, the patient is watched for pain, fever and emphysema, and follow-up endoscopy or a contrast study is arranged if a stricture or tumour is suspected.

4 CHAPTER FOUR Special objects

Batteries, magnets, sharps.

4.1 Button (disc) batteries

A button battery lodged against moist mucosa passes a current that *hydrolyses tissue fluid and generates hydroxide at the negative pole*, producing an alkaline, liquefactive burn, to which leakage of the contents and pressure add. The injury is fast: the mucosa is damaged within an hour, the muscle within two to four hours, and the wall can perforate within 8 to 12 hours. Lithium cells of 20 mm or more are the most dangerous. Late complications include tracheo-oesophageal fistula, mediastinitis, stricture, vocal fold palsy and an *aorto-oesophageal fistula with fatal haemorrhage*, which can appear days or weeks later.

A battery in the oesophagus is removed *endoscopically at once*, at any hour. If ingestion was within 12 hours, honey (in children over one year) or sucralfate is given on the way to theatre to coat and protect the mucosa, provided it does not delay removal. Afterwards CT or MRI looks for deep injury, and significant injuries are observed in hospital.

Once past the oesophagus, most batteries in the stomach pass. They are removed if the child has symptoms, if a large battery has been swallowed by a young child, or if it remains in the stomach on a repeat film at about 48 hours; Dhingra's threshold is a child under six with a battery of 1.5 cm or more. Batteries in the nose and ear are emergencies too.

THE HALO SIGN

Any round radio-opaque object in the oesophagus of a child is a button battery until a careful look excludes it. Look for the double contour (halo) on the frontal film and the step at the edge on the lateral. Remove it now, not on tomorrow's list.

4.2 Magnets

A single magnet usually passes. Two or more magnets, or a magnet with a metal object, attract each other across loops of bowel and cause pressure necrosis, fistulae and perforation. Remove them while they can be reached endoscopically, and seek early surgical advice if they have passed beyond.

4.3 Sharp and long objects

Pins, needles, bones, open safety pins and denture fragments with hooks carry the highest risk of perforation and are removed from the oesophagus urgently. Beyond the stomach, sharp objects and long objects (over about 5 cm in a small child and 6 cm or more in adults) may not negotiate the duodenum; they are removed endoscopically if reachable, or followed with daily films, and surgery is needed for pain, tenderness, failure to progress or perforation.

4.4 Objects beyond the stomach

Most objects that reach the stomach pass within a week on a normal diet, without purgatives, while the stools are checked and films repeated. Operation is needed for abdominal pain or tenderness, an object that does not progress on serial films, a sharp object likely to perforate, a long object in a small child, and pyloric stenosis or other obstruction.

4.5 Food bolus impaction

If the patient can swallow saliva and is comfortable, a short wait may allow the bolus to pass. If saliva cannot be swallowed, or impaction lasts beyond 12 to 24 hours, the bolus is removed endoscopically, gently pushed on if there is no stricture or removed piecemeal. Almost every adult then needs investigation, usually *biopsy for eosinophilic oesophagitis* and a search for rings, strictures and cancer.

5 CHAPTER FIVE Complications

Why we do not wait.

5.1 What a neglected object does

A foreign body can obstruct the airway through laryngeal oedema or, in children, tracheal compression. Perforation leads to *retropharyngeal or parapharyngeal abscess* and *mediastinitis*, and a sharp object can erode into the *aorta* to form an aorto-oesophageal fistula, announced by a sentinel bleed before exsanguination; pericarditis and empyema can also follow. Periesophageal cellulitis and neck abscess, tracheo-oesophageal fistula, ulceration and stricture from a neglected object, and migration of needles and bones into the neck, sometimes needing open exploration, complete the list.

PHARYNGEAL AND OESOPHAGEAL FOREIGN BODIES AT A GLANCE

SITUATION	ACTION
Fish bone seen in tonsil or tongue base	Remove under topical anaesthesia with forceps

SITUATION	ACTION
Persistent pricking with a normal examination	Flexible endoscopy; CT if still suspected; oesophagoscopy if seen or symptoms persist
Coin in the upper oesophagus	Rigid or flexible removal within 24 hours
Coin in the lower oesophagus, well child	Observe up to 24 hours; repeat film; remove if still there
Button battery in the oesophagus	Immediate removal; honey or sucralfate en route if it does not delay
Food bolus, cannot swallow saliva	Urgent endoscopy; biopsy for eosinophilic oesophagitis
Sharp object in the oesophagus	Urgent rigid oesophagoscopy, sheathing the point
Two or more magnets	Remove while reachable; surgical review
Pain, fever, emphysema	Suspect perforation: nil by mouth, antibiotics, CT, surgical opinion

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Commonest site** (just below the cricopharyngeus) and the three narrowings.
- **Coin orientation:** oesophagus versus trachea.
- **Button battery:** halo sign, mechanism and timing of injury, immediate removal, honey or sucralfate.
- **Signs of an oesophageal foreign body** and of perforation.
- **Radiolucent objects;** why barium is avoided; the role of CT.
- **Rigid oesophagoscopy technique** for sharp objects.

- **What not to do:** Foley catheter, pushing down, papain.

6.2 Common pitfalls

COMMON PITFALLS

- Mistaking a button battery for a coin.
- Sending home a patient with a persistent foreign body sensation and a "normal X-ray".
- Using barium as the first study.
- Forgetting to look for an underlying stricture, eosinophilic oesophagitis or cancer in an adult with bolus impaction.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, Oxford Specialist Handbook ch. 10, Oxford Handbook of ENT ch. 25, Dhingra ch. 70 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…